

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

Iris Analysis

```
# Assign the iris dataset to a new variable called "my_iris"
my_iris <- iris
```

Average Sepal Length by Species

```
# Assuming your dataset is called "iris"
my_iris %>%
  group_by(Species) %>%
  summarize(mean_sepal_length = mean(Sepal.Length))
```

```
## # A tibble: 3 x 2
##   Species    mean_sepal_length
##   <fct>          <dbl>
## 1 setosa          5.01
## 2 versicolor     5.94
## 3 virginica       6.59
```

Visualizations

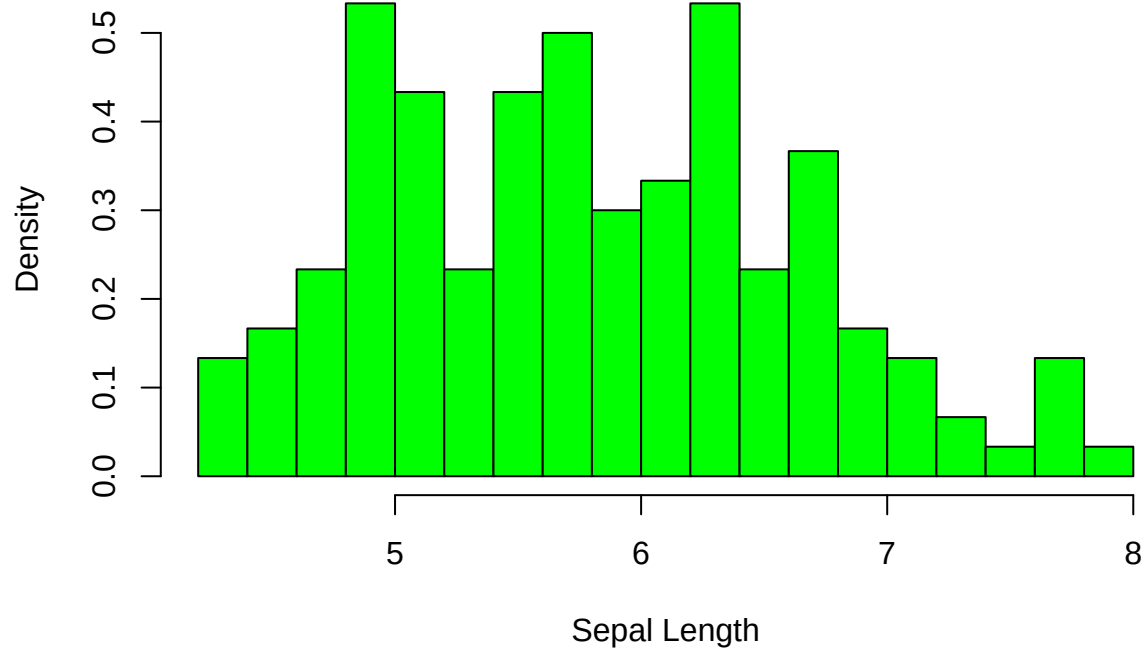
```
# Printing top 5 rows
head(my_iris,5)
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1         5.1         3.5         1.4         0.2  setosa
## 2         4.9         3.0         1.4         0.2  setosa
## 3         4.7         3.2         1.3         0.2  setosa
## 4         4.6         3.1         1.5         0.2  setosa
## 5         5.0         3.6         1.4         0.2  setosa
```

```
sepal_length<-my_iris$Sepal.Length
sepal_width<-my_iris$Sepal.Width
```

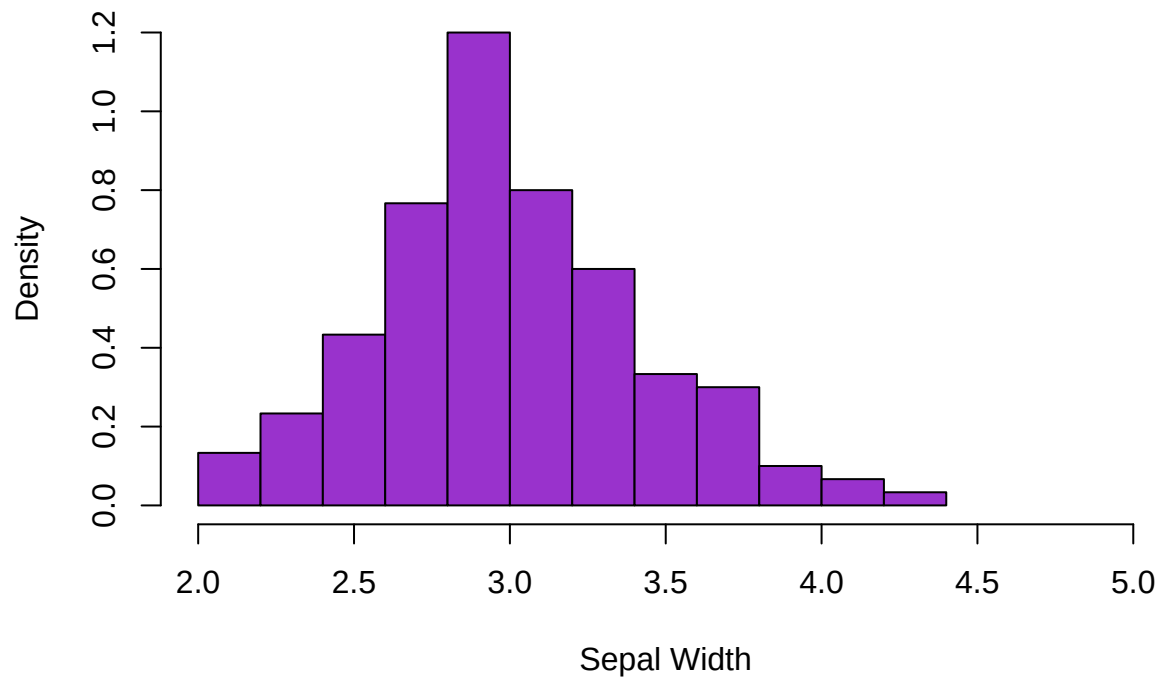
```
hist(sepal_length,main="Histogram - Sepal Length",xlab="Sepal Length"
     ,breaks = 20, col="green",freq=FALSE)
```

Histogram – Sepal Length

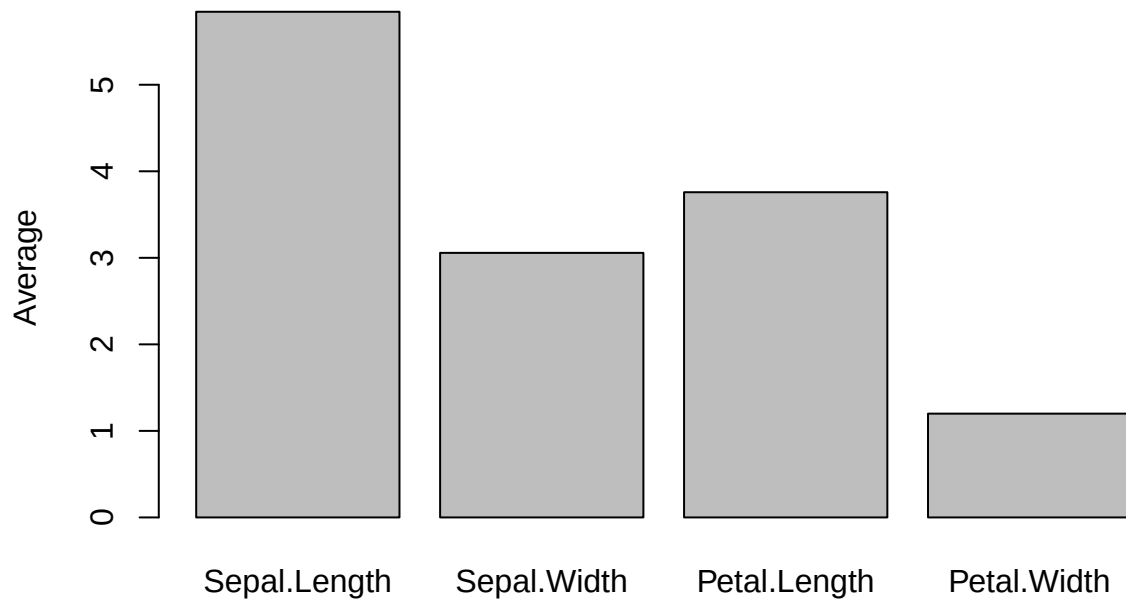


```
hist(sepal_width,main="Histogram – Sepal Width",xlab="Sepal Width",xlim=c(2,5), col="darkorchid",freq=F)
```

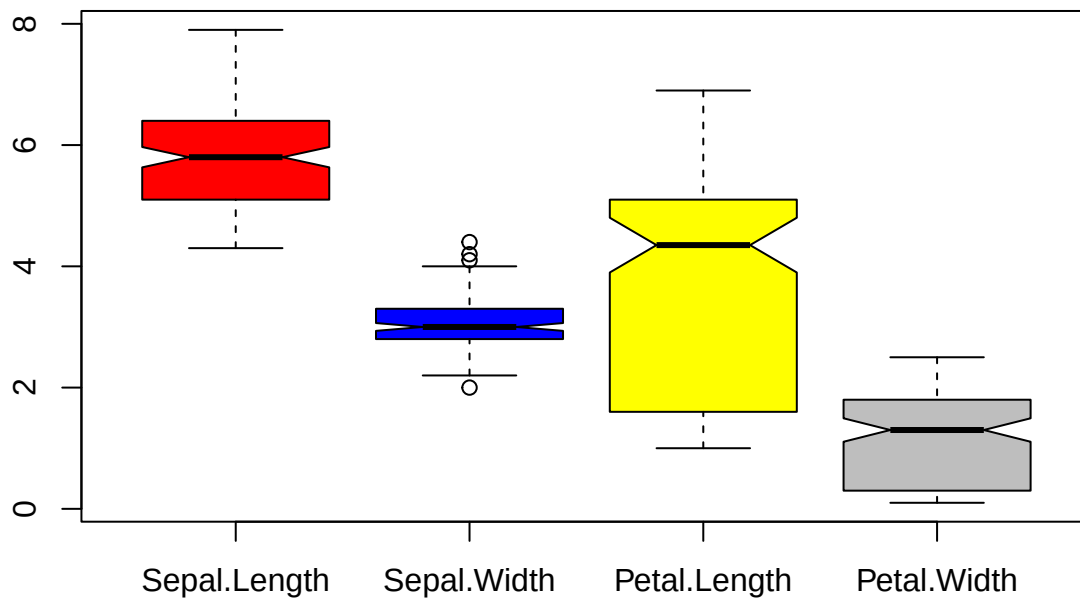
Histogram – Sepal Width



```
# bar chart for average of the 4 quantitative variables
aveg<- apply(my_iris[,1:4], 2, mean)
barplot(aveg, ylab = "Average")
```



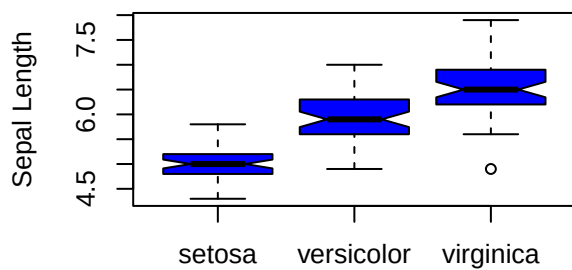
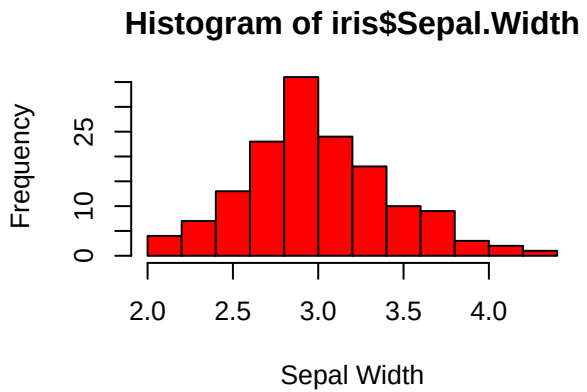
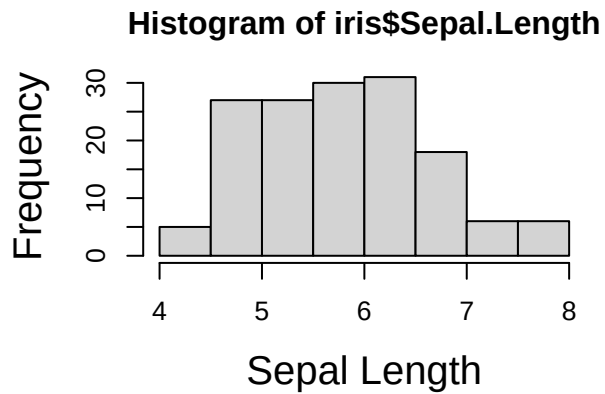
```
## box plot of multiple variables into one figure
boxplot(iris[,1:4], notch=T, col=c("red", "blue", "yellow", "grey"))
```



```
# set arrangement of multiple plots
par(mfrow=c(2,2))

# set margins
par(mar=c(4.5,4.2,3,1.5))
hist(iris$Sepal.Length,xlab = "Sepal Length",cex.lab=1.5)
hist(iris$Sepal.Width,xlab = "Sepal Width",col="red")

boxplot(iris[,1]~iris[,5], notch=T,ylab="Sepal Length",col="blue")
```



iris[, 5]

Here is analysis based on Visualization for different Species: ### The three Iris species differ in their petal dimensions, with “setosa” ### significantly shorter petals compared to both “versicolor” and “virginica”. #

KEY POINTS:-

- 1.”versicolor” has slightly longer petals than “setosa”.
- 2.”virginica” stands out with notably longer and wider petals.
- 3.”setosa” shortest petals.
- 4.”versicolor” moderate long petals intermediate b/w “setosa” & “virginica”
- 5.”virginica”: longest and widest petals.